

Three-Band Graphic Audio Equalizer Project Report

1. Introduction

A band-pass filter (BPF) is a circuit that allows signals within a specific frequency range to pass while attenuating frequencies outside this range. In audio applications, band-pass filters are widely used in equalizers, communication systems, and instrumentation to isolate or emphasize certain frequency bands.

For this project, a three-band graphic equalizer is designed using the Deliyannis–Friend multiple-feedback topology, targeting 60 Hz, 280 Hz, and 1200 Hz. This topology is chosen for its ability to provide narrow bandwidth control, stable performance, and implementation without inductors.

2. Theory Concepts

2.1 Band-Pass Filter Basics

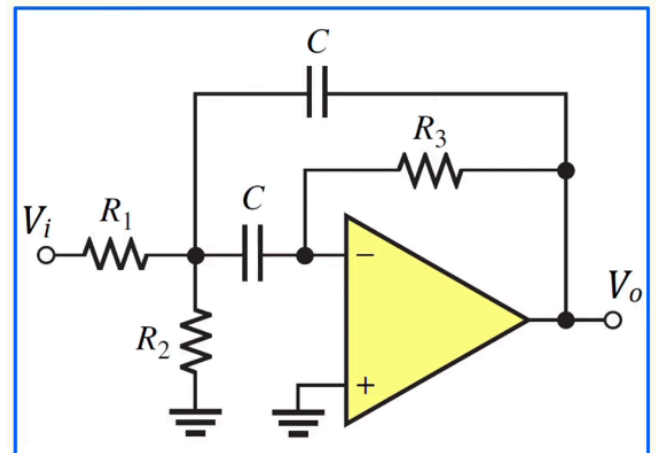
A band-pass filter is defined by three parameters:

- Center Frequency (f_o): Frequency at which the filter response peaks.
- Bandwidth (BW): Frequency range between the **-3 dB**(half power) cutoff points.
- Quality Factor (Q): Measure of selectivity,

defined as
$$Q = \frac{f_o}{BW}$$

2.2 Deliyannis–Friend Band-Pass Filter

The Deliyannis–Friend filter is a modified multiple-feedback band-pass filter. It uses one operational amplifier, resistors, and capacitors to realize second-order band-pass characteristics with controllable gain and Q. It is a second order band pass filter as it has two storage elements, capacitors of the same value.



Its Transfer Function is given by

$$H(s) = \frac{V_o(s)}{V_i(s)} = \frac{-\frac{1}{R_1 C} s}{s^2 + \frac{2}{R_3 C} s + \frac{R_1 + R_2}{R_1 R_2 R_3 C^2}}$$

The Standard Form of Second Order Bandpass Filter:

$$H(s) = \frac{K \omega_{BW} s}{s^2 + \omega_{BW} s + \omega_{center}^2}$$

Comparing Terms:

$$K \omega_{BW} = \frac{1}{R_1 C} \quad \omega_{BW} = \frac{2}{R_3 C} \quad \omega_{center} = \frac{1}{\sqrt{(R_1 \parallel R_2) R_3 C}}$$

$$\text{Where } R_1 \parallel R_2 = \frac{R_1 \times R_2}{R_1 + R_2}$$

The Design equations are:

$$R_3 = \frac{2Q}{\omega_{center} C} \quad R_2 = \frac{R_3}{4Q^2 - 2K} \quad R_1 = \frac{R_3}{2K} \quad Q = \frac{\omega_{center}}{\omega_{BW}}$$

This filter is widely used in audio equalization because it allows stable implementation of high-Q band-pass responses without inductors.

3. Calculation of Resistance Values

Given design targets:

- Band 1

$$f_0 = 60 \text{ Hz} \quad BW = 80 \text{ Hz}$$

- Band 2

$$f_0 = 280 \text{ Hz} \quad BW = 370 \text{ Hz}$$

- Band 3

$$f_0 = 1200 \text{ Hz} \quad BW = 1470 \text{ Hz}$$

Here the common quality factor stands around $Q=0.75$

The capacitor value is taken as $c=0.1$ micro farads

Parameter	Filter 1	Filter 2	Filter 3
Center Frequency (f_o)	60 Hz	280 Hz	1200 Hz
Bandwidth (BW)	80 Hz	370 Hz	1470 Hz
Quality Factor (Q_p)	0.750	0.757	0.816
Resistor R1	19.9 k Ω	4.30 k Ω	1.08 k Ω
Resistor R2	39.8 k Ω	8.61 k Ω	2.16 k Ω
Resistor R3	159.2 k Ω	29.48 k Ω	3.27 k Ω

4. Working of the Circuit

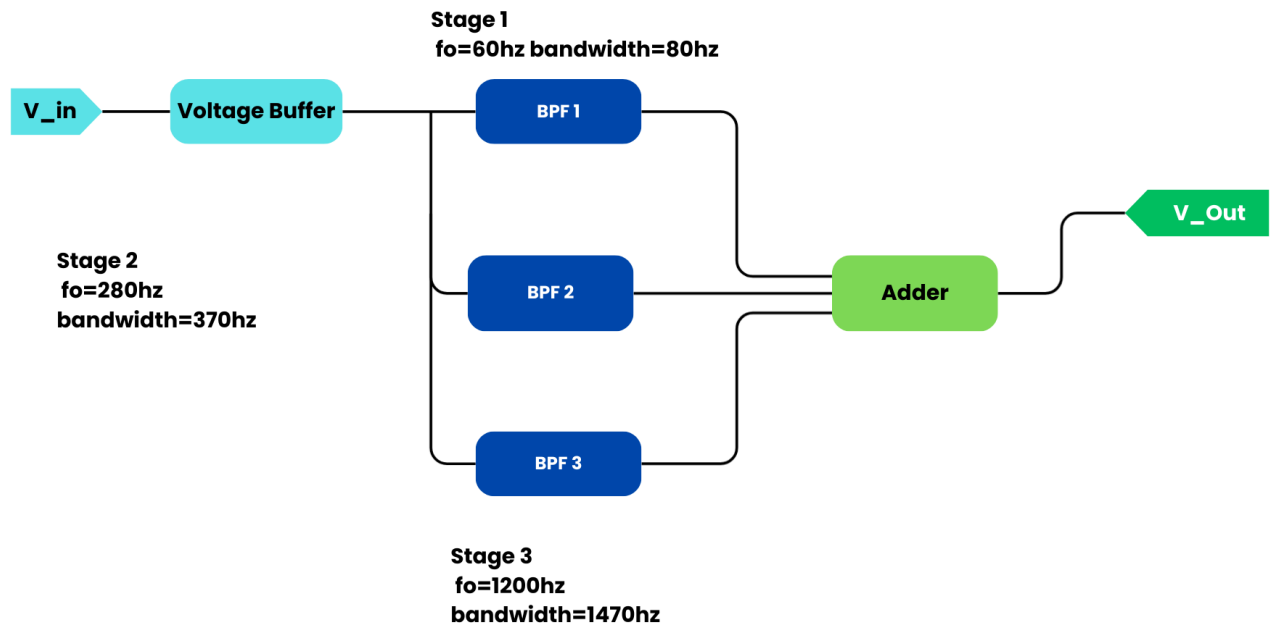
1. Input Buffer: The audio input is first fed through a buffer to prevent loading effects.
2. Band-Pass Stages: Each of the three Deliyannis–Friend band-pass filters processes a specific frequency range (low, mid, high).
3. Parallel Combination: The outputs of the three filters are adjusted using potentiometers (gain control for each band).
4. Summing Amplifier: The filtered signals are added together to form the equalized output.
5. Output Stage: Final signal is sent to speakers or acquisition system (ADALM1000 in this case).

Thus, by adjusting the potentiometers, the user can boost or cut different audio frequency bands, effectively shaping the sound spectrum.

5. Block Diagram

The system is structured into the following blocks:

- Audio Input Source → Buffer Amplifier
- Three Parallel Deliyannis–Friend Band-Pass Filters (60 Hz, 280 Hz, 1200 Hz)
- Summing Amplifier → Composite Output
- ADALM1000 Interface → MATLAB Visualization



6. Experimental Workflow and Validation

- **Op-Amps:** Low-noise ICs (MCP6004, LM324) used for audio quality.
- **Passive Components:** 1% metal-film resistors and stable polyester capacitors.
- **Potentiometers:** Added to control gain for each band.
- **Breadboard Prototype:** Layout designed to reduce crosstalk, decoupling capacitors for supply stability.

6.1 Initial Music Input Testing

The first phase of experimentation involved feeding a music WAV file into the graphic equalizer circuit. The WAV file was played back from MATLAB, and the processed audio output was sampled via the ADALM1000 data acquisition system. The default sampling rate of the ADALM1000 hardware is 10 kHz, which is insufficient for full-range audio signals that require at least 44.1 kHz sampling to ensure fidelity and prevent aliasing. To

address this, the driver code was modified to enable higher-rate sampling in MATLAB, allowing the system to operate at the required standard for audio experiments.

Despite these improvements, distortions were observed in the output signal. Key contributors to this distortion included:

- **Speaker limitations:** Nonlinearities and limited bandwidth affected the output waveform. The sampled output audio had inconsistencies which created distortions in the sound.
- **ADALM1000 driver and firmware constraints:** While sampling rate was increased, real-time streaming and hardware limitations introduced quantization and data transfer errors as the audio devices and the ideal sampling frequency of the kit were mismatched so the sampling took place at 44.1 kHz.
- **Power supply limitations:** Inadequate supply voltage, which limited the gain to at max 3 times the input voltage

These issues complicated the analysis of complex music signals, making it difficult to interpret filter performance using spectral plots alone.

6.2 Targeted Sinusoidal Input

Based on guidance from faculty, the next phase transitioned to a more controlled experimental input: a composite signal formed by adding together several sine waves at distinct frequencies, each corresponding to the center frequency of one of the three bandpass filters (e.g., 60 Hz, 280 Hz, and 1200 Hz). This approach has multiple advantages:

- The output could be more precisely analyzed, as each filter's passband would uniquely isolate its corresponding frequency component.
- Verification became direct and clear: Each filter's output was expected to show a strong response only when its associated input frequency was present.
- Reduces ambiguity from music's complex frequency content, providing clear diagnostic peaks in the time and frequency domains.

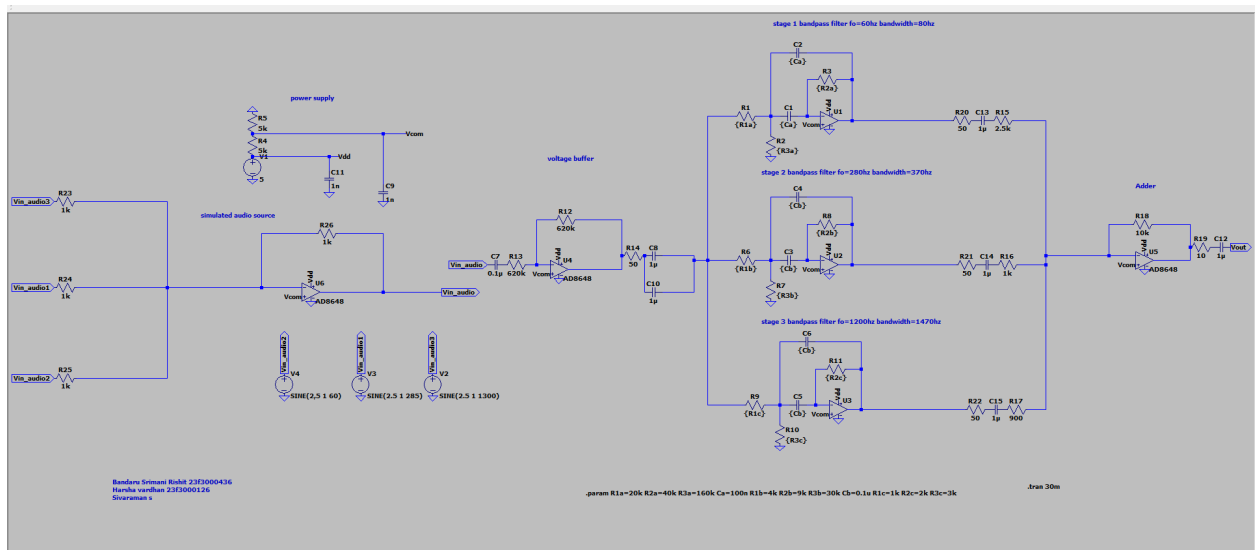
By carefully selecting the frequencies of the input sine waves and monitoring the circuit's output—using both the **ADALM1000 DAQ** and MATLAB's plotting and FFT features—the effectiveness and selectivity of each bandpass filter stage were validated.

During testing, the gain of two filter stages was held constant while adjusting the third to unity, enabling selective amplification and direct measurement of each band's frequency behavior. Due to the output voltage constraint of 5 V, the maximum gain for all filters was limited to 3. This stepwise procedure allowed reliable assessment of band-specific amplification without clipping or nonlinear distortion, as well as systematic testing of frequency selectivity and filter performance under practical voltage constraints.

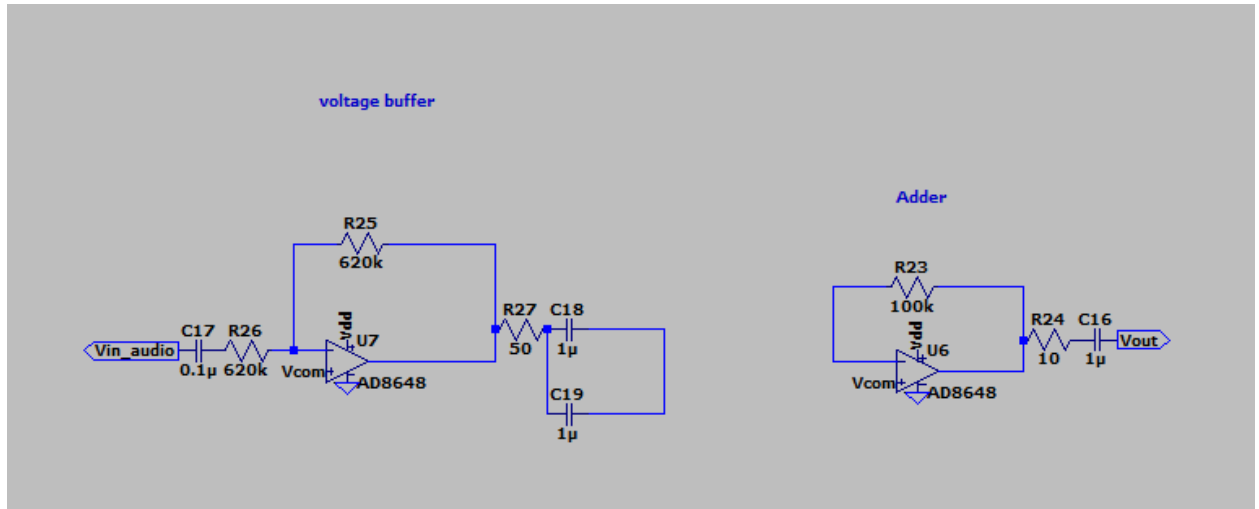
7. Software Implementation

7.1 Simulation and Software Verification

- LTSpice was used to simulate the analog equalizer circuit, producing reference Bode plots for each band, which confirmed proper center frequency and bandwidth selection in the initial design.



- Figure 1. Simulation 3 Band Graphic Equaliser in LTSpice



- *Figure 2. Voltage Buffer & Adder Circuit*

- MATLAB models of digital Butterworth bandpass filters were created in parallel, generating FFTs and frequency response plots to serve as theoretical benchmarks.

Comparison between LTSpice, MATLAB, and eventual experimental results provided comprehensive verification of frequency selectivity and spectral shaping.

7.2 Hardware Data Acquisition and Processing

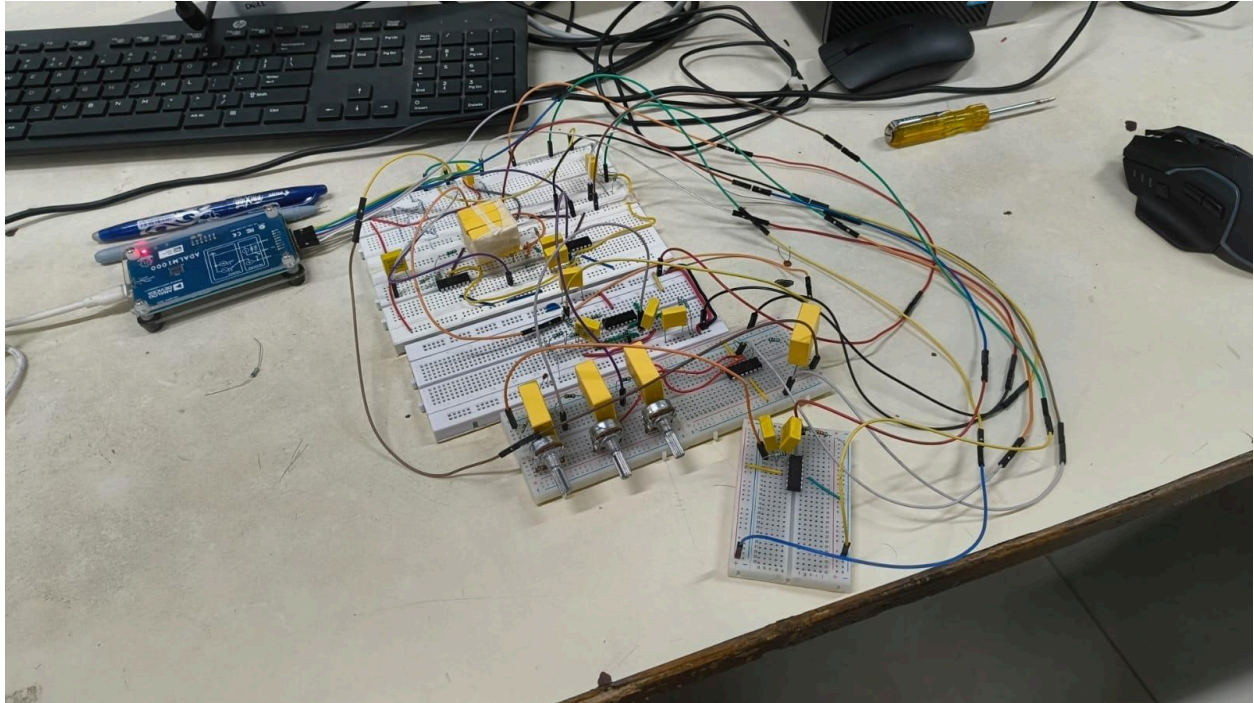
The breadboarded circuit output was connected to the **ADALM1000 DAQ system**. Signals were streamed to MATLAB, where post-processing included offset removal, normalization, and spectral analysis.

This workflow enabled direct, quantitative comparison between simulated and measured hardware responses, including real-time validation during adjustments.

8. Experimental and Comparative Results

8.1 Circuit Schematic

We have implemented 3 stage D-F bandpass filters with 3 different center frequencies, the circuit also included the voltage buffer in input stage, and an adder in output stage

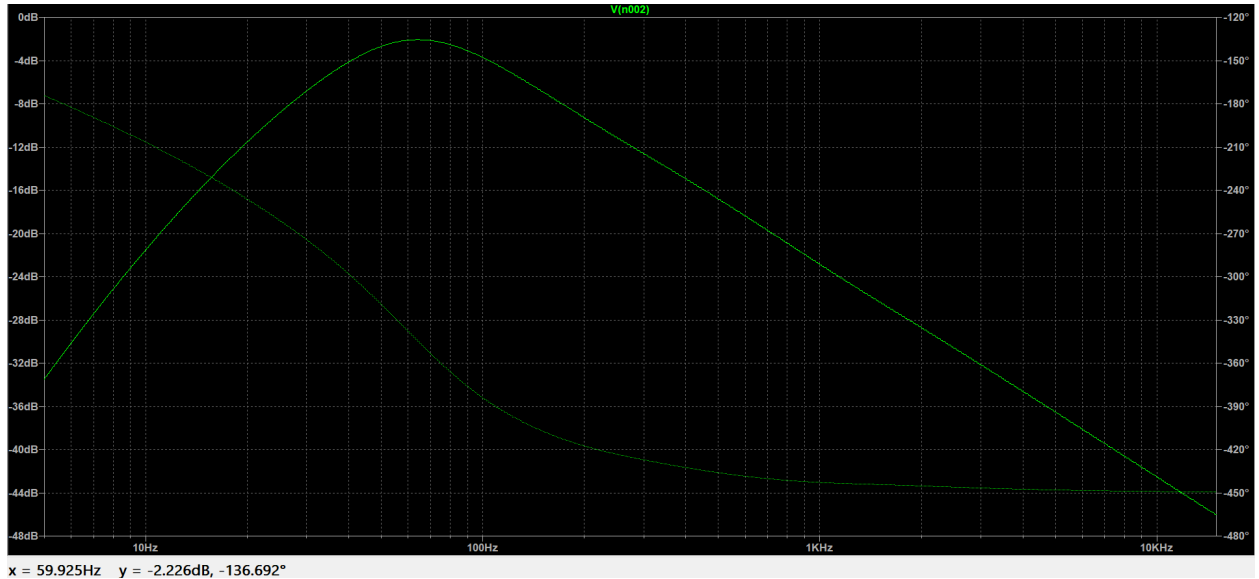


- *Figure 1. Graphic Equaliser implemented on breadboard*

8.2 Simulation results (LTSpice)

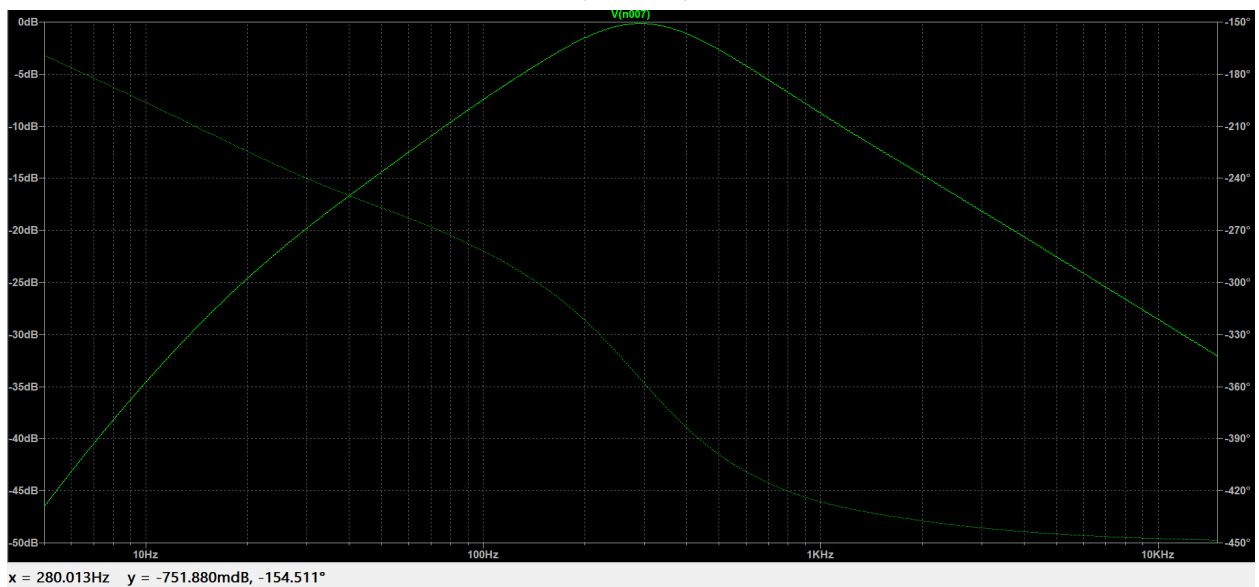
To validate the design, Bode plots were first generated using LTSpice simulation for each bandpass stage. Each simulated response confirms the intended center frequency and bandwidth.

- *Figure 1. Simulated Bode Plot for Stage 1 (60 Hz)*



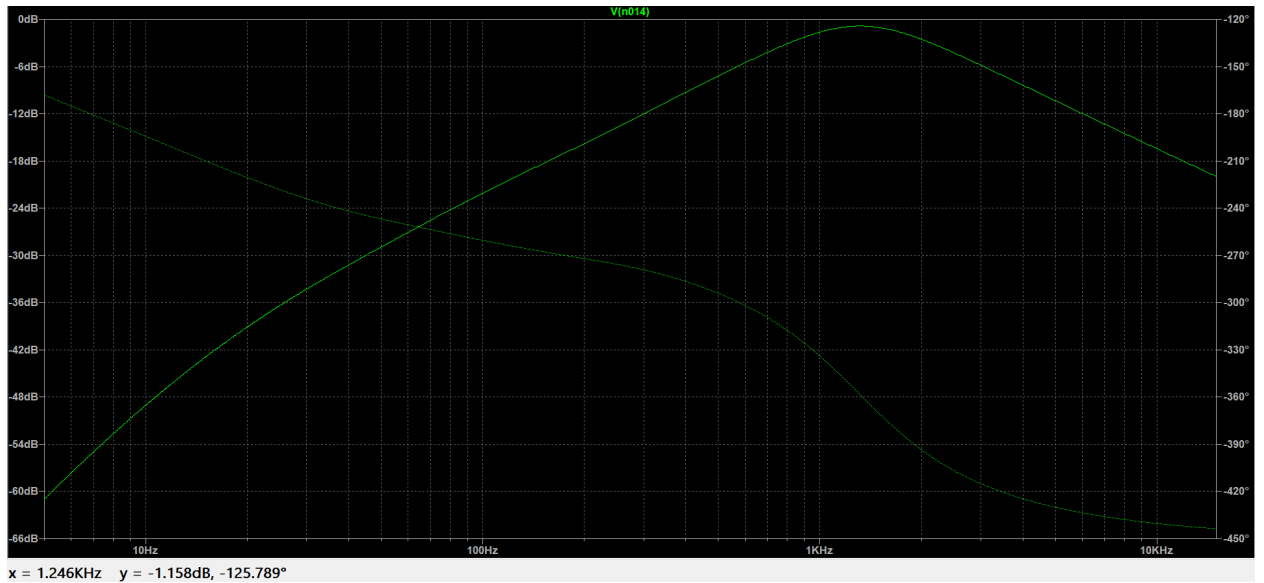
The peak of the bode plot is around the centre frequency of 60Hz.

- *Figure 3. Simulated Bode Plot for Stage 2 (280 Hz)*



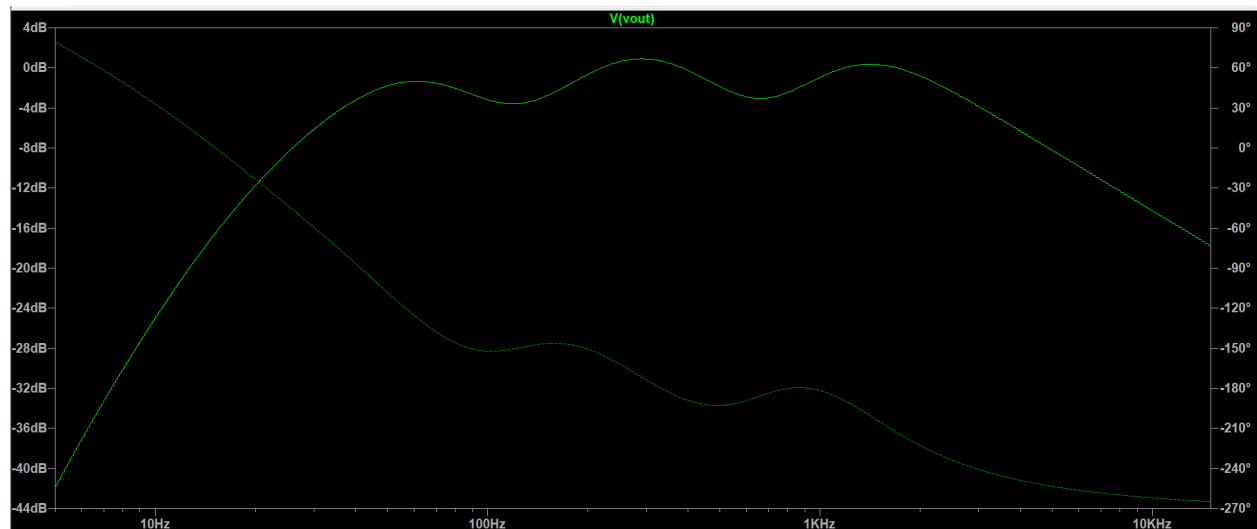
The peak of the bode plot is around the centre frequency of 280Hz.

- *Figure 3. Simulated Bode Plot for Stage 3 (1200 Hz)*



The peak of the bode plot is around the centre frequency of 1200Hz.

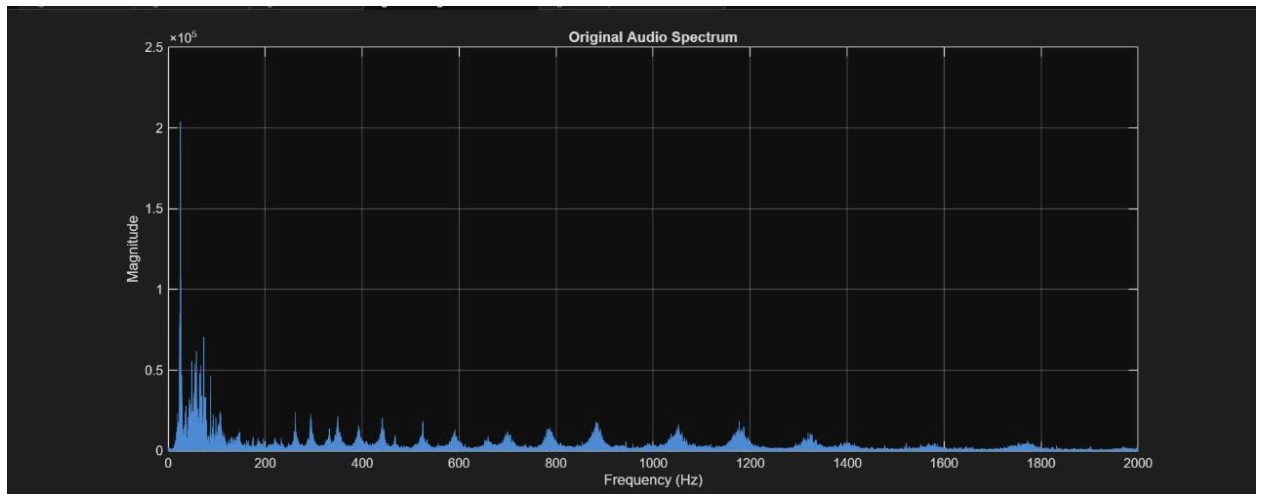
Figure 4: final circuit bode plot showing the frequency response of the circuit



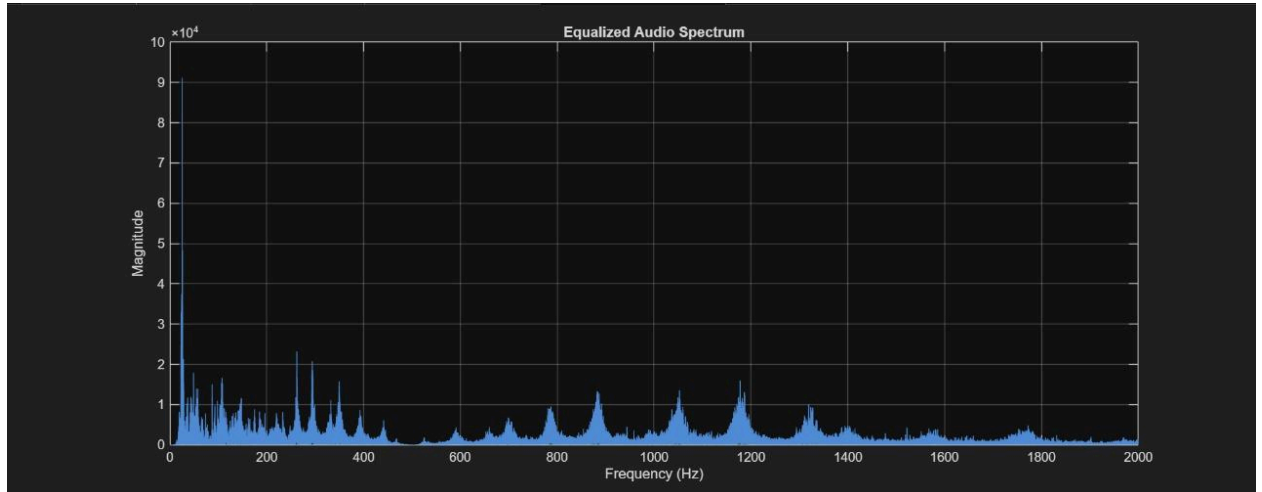
8.3 MATLAB digital filter simulations

Spectral analysis was performed comparing the original, simulated, and equalized audio using FFT:

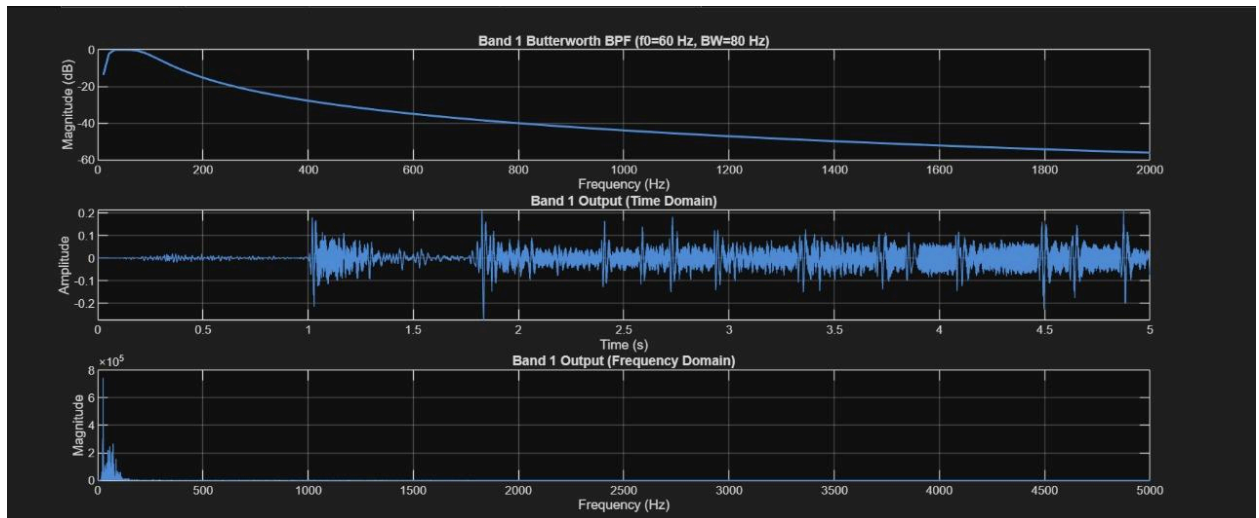
- *Figure 8. FFT of Original Audio (Input Signal)*



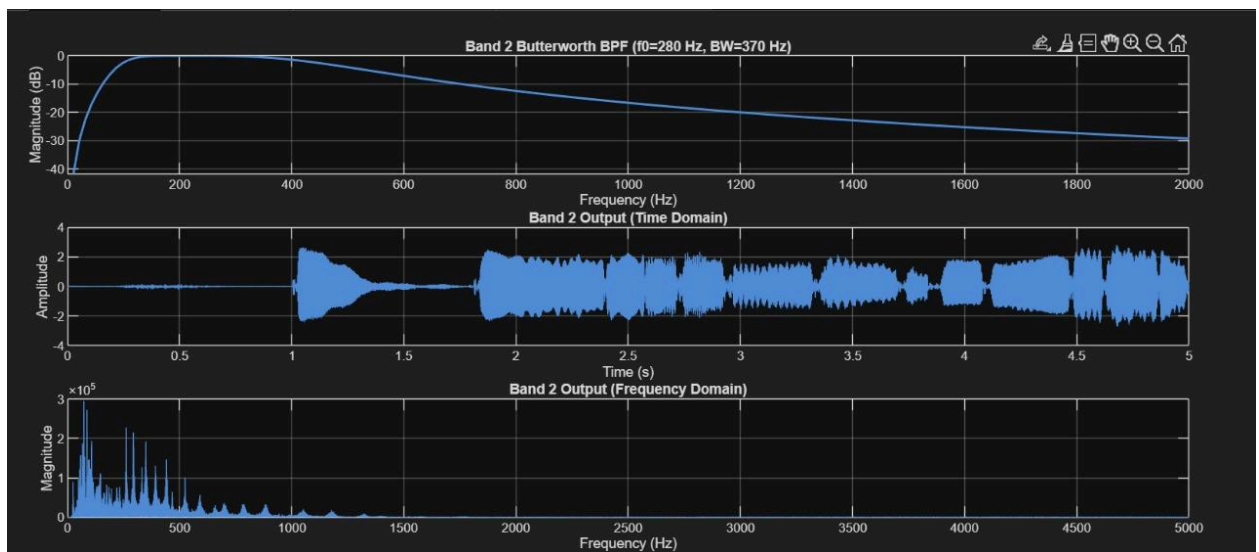
- *Figure 9. FFT of Equalized Audio (Output Signal)*



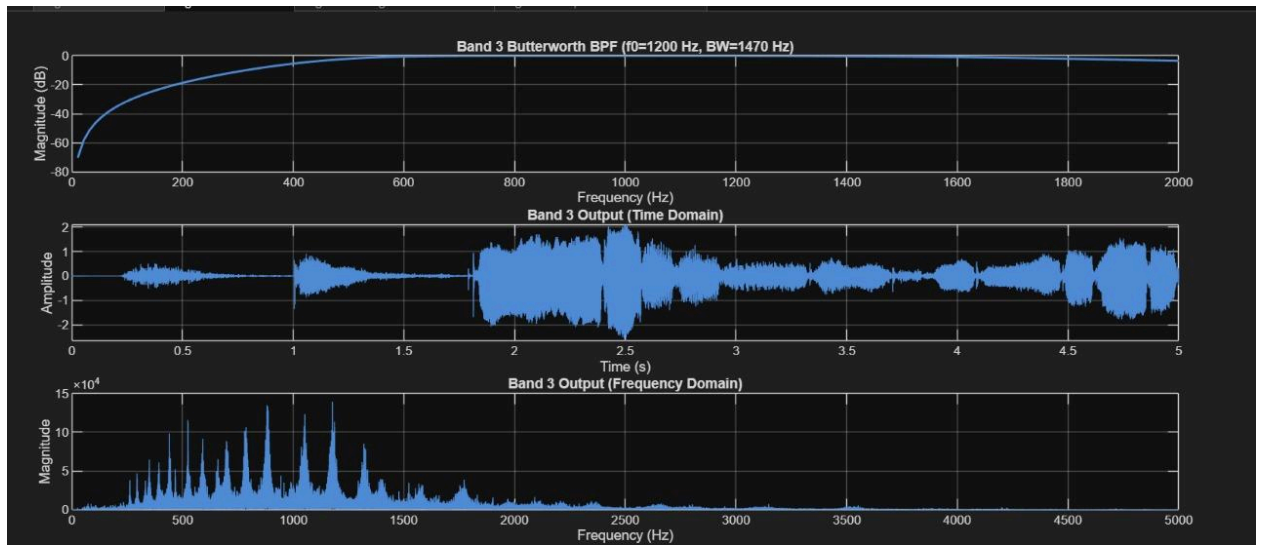
- *Figure 10. Digital Filter Simulation (MATLAB) for Stage 1 (60 Hz)*



- *Figure 11. Digital Filter Simulation (MATLAB) for Stage 2 (280 Hz)*



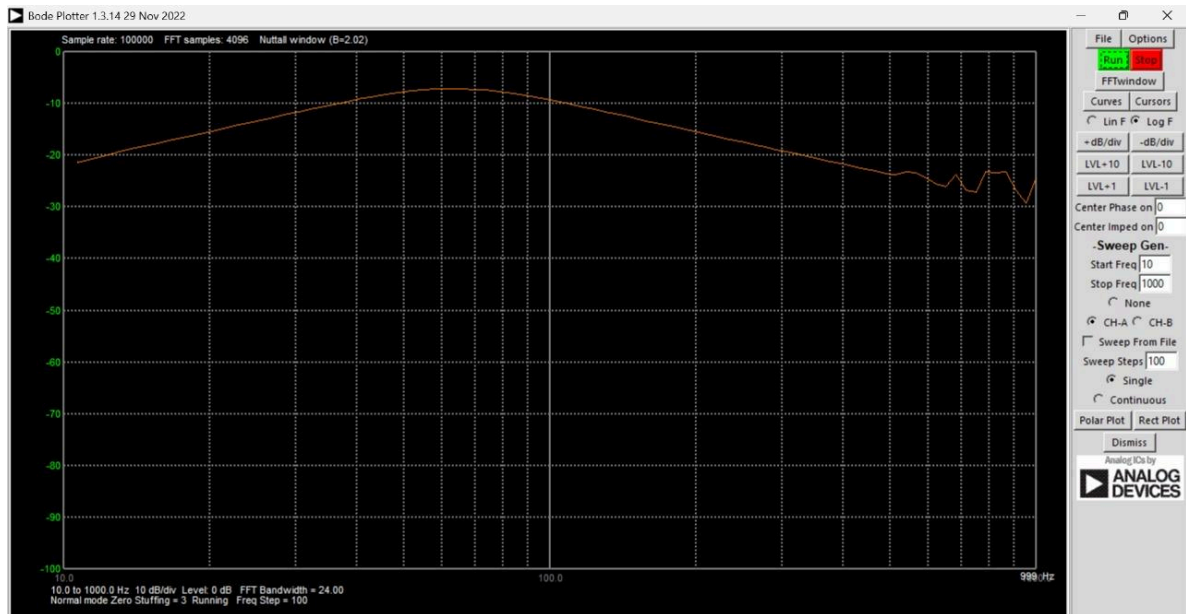
- *Figure 12. Digital Filter Simulation (MATLAB) for Stage 3 (1200 Hz)*



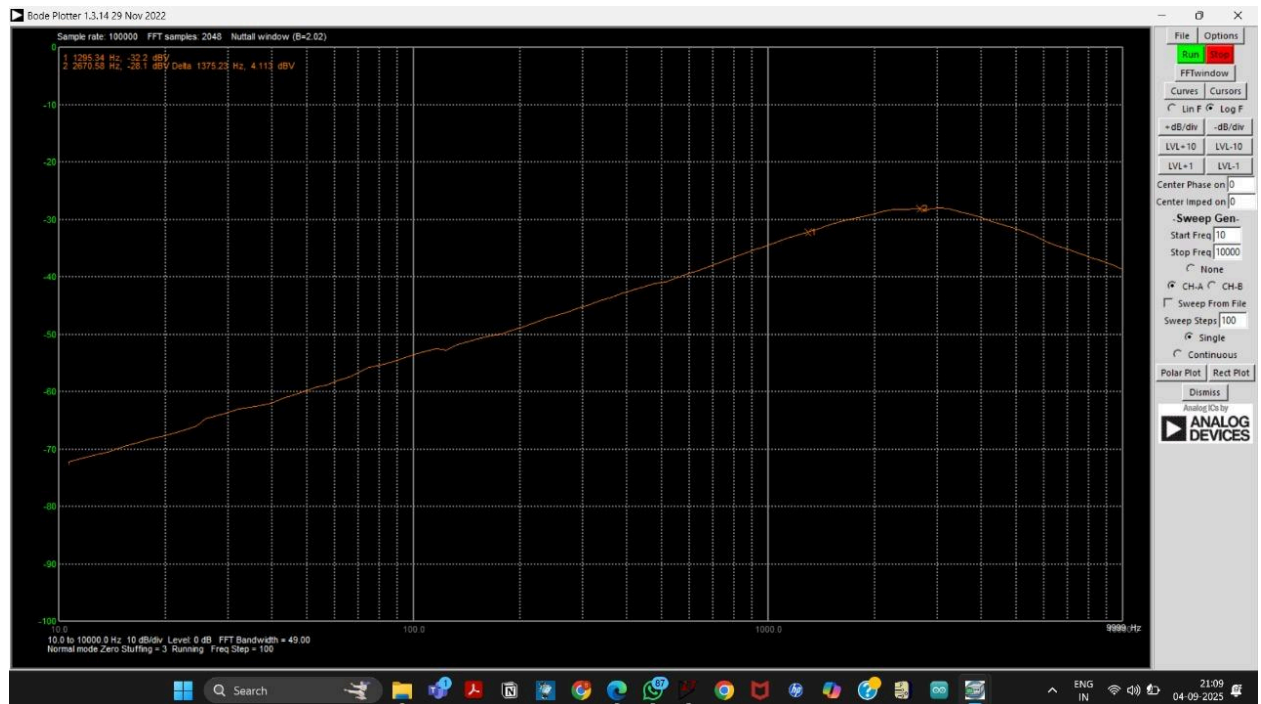
8.3 Hardware measurement results (Bode plots, FFT from DAQ)

After hardware assembly, the circuit was tested using ADALM1000 and MATLAB for real-time data capture. The measured Bode plots below represent the actual gain response for each band in the physical circuit.

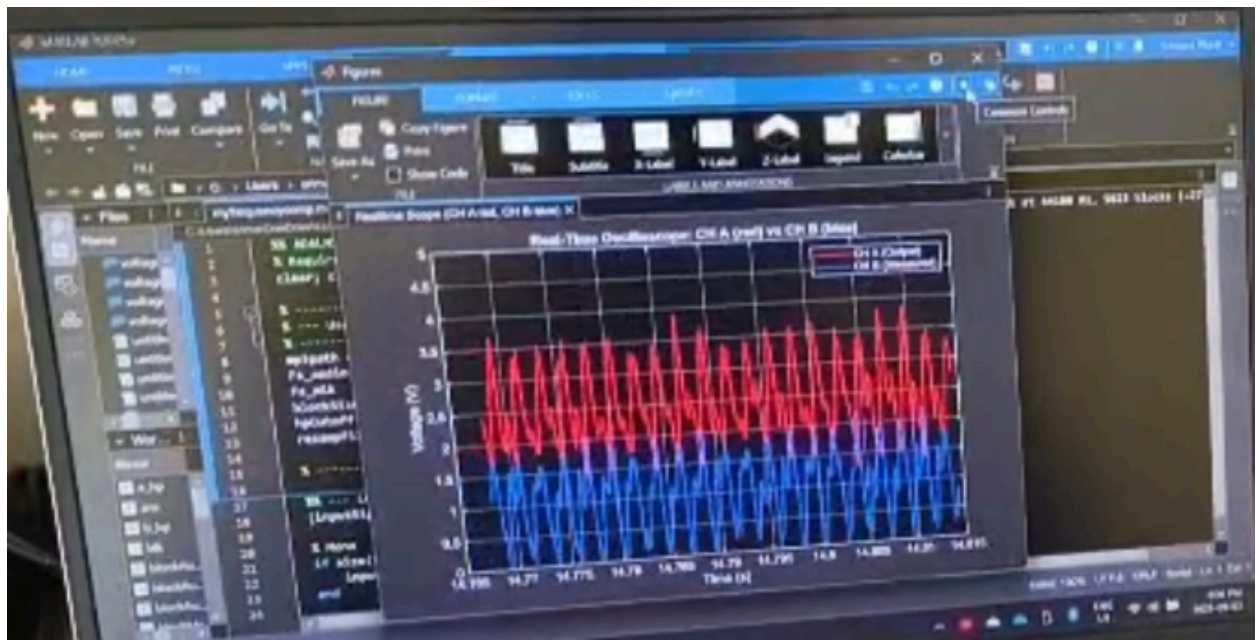
- *Figure 4. Experimental Bode Plot for Stage 1 (60 Hz)*



- Figure 5. Experimental Bode Plot for Stage 3 (1200 Hz)



9. Audio output



10. Discussion

- **Analog vs. Digital:** MATLAB filters replicate the design but lack real-world parasitics.
 - **Component Tolerances:** Slight deviations observed in hardware peak frequencies.
 - **Op-Amp Choice:** High-quality op-amps reduce distortion and noise.
 - **Scalability:** Same design principle can extend to 5-band or 10-band equalizers.
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11. Conclusion

The project successfully designed, simulated, and implemented a three-band graphic audio equalizer using the Deliyannis–Friend band-pass topology. Hardware results match theoretical predictions and simulations, with minor differences explained by practical factors. The use of ADALM1000 for real-time acquisition enhanced validation, confirming the equalizer’s ability to shape audio signals effectively.

12. References

1. Texas Instruments, *An audio circuit collection, Part 3*, SLYT134.
https://www.ti.com/lit/an/slyt134/slyt134.pdf?ts=1755593905383&ref_url=https%253A%252F%252Fwww.google.com%252F
2. MATLAB Documentation, Data Acquisition Toolbox.

